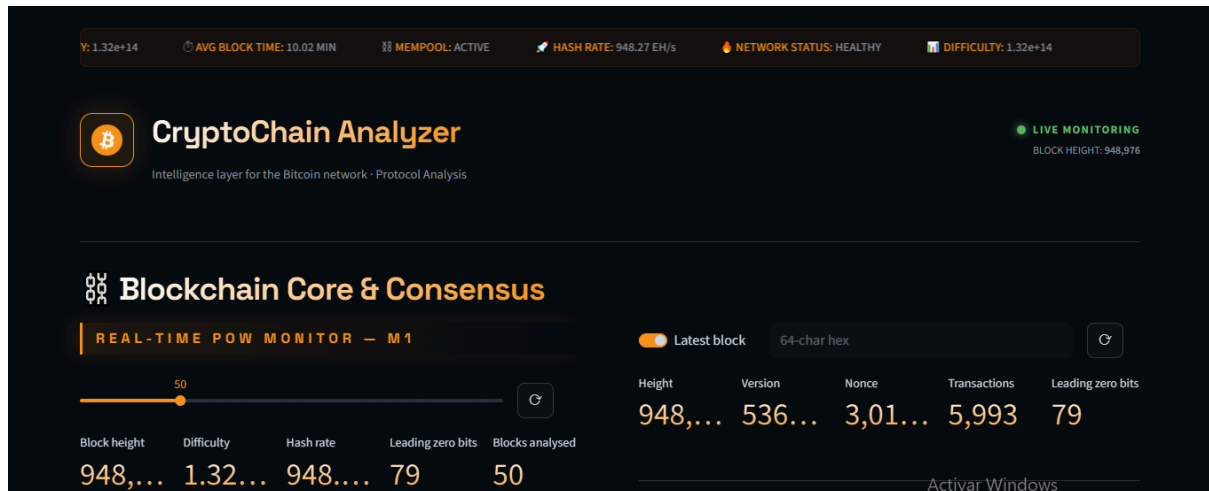


CryptoChain Analyzer



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Course: Cryptography | Universidad Alfonso X el Sabio

1. Cryptographic Metrics and Meaning The dashboard features several modules that connect theoretical cryptographic concepts with real-time data from the Bitcoin network.

- **Proof of Work and Target Threshold:** Bitcoin's security is centered on the Proof of Work (PoW) mechanism. In module M2, we verify blocks by analyzing their headers and ensuring the resulting hash is below the network's specific target. This demonstrates how difficulty adjustment works: as mining power fluctuates, the target scales to keep block production at 10-minute intervals.
- **Difficulty History and Adjustment:** Module M3 tracks how difficulty evolves over time. Bitcoin adjusts this difficulty every 2016 blocks based on the ratio between the actual time taken and the 2-week target.

2. AI Component: Statistical Anomaly Detection In module M4, we implemented an anomaly detector based on the time intervals between blocks.

- **Model Selection:** Since mining follows a Poisson process, the time between blocks follows an exponential distribution. We use statistical calculations (MLE) to fit the model and identify outliers.
- **Evaluation:** The model is tested using the Kolmogorov-Smirnov test to ensure the theoretical model accurately reflects real-world data.

3. Beyond Course Notes: Security and Network Analysis * 51%

Attack Cost: We estimate the hardware and electricity costs required to overtake the network, which currently requires billions of dollars in infrastructure.

- **Nakamoto Double-Spend Probability:** We visualize the math from the original 2008 whitepaper regarding the probability of an attacker catching up after a certain number of confirmations.
- **Mempool & Fee Market:** Real-time monitoring of network congestion and recommended transaction fees.
- **Scalability and SegWit:** An analysis of how SegWit enables "Layer 2" scaling by separating digital signatures from transaction data.
- **Nonce Entropy & ASIC Fingerprinting:** Statistical analysis of the random numbers (nonces) used by miners to identify patterns in hardware behavior.

4. UI/UX: Hypermodern Cinematic Design The dashboard uses a "Hypermodern" aesthetic inspired by high-end data visualization:

- **Perspective Grid Drift:** A moving background that creates a sense of depth and data flow.
- **Cinematic Textures:** Subtle film grain for a professional, high-end feel.
- **Holographic Gradients:** Multi-tone colors that simulate high-tech displays.
- **Floating Glassmorphism:** Components that appear to float with deep background blurs.

- **Micro-interactions:** Interactive scaling and neon pulse effects to engage the user.

5. References * Satoshi Nakamoto (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.

- API Documentation from Blockstream.info and Mempool.space.
- Universidad Alfonso X el Sabio (Topic 7: Blockchain and Cryptography notes).